

This is a document for describing and proposing what we would like to accomplish with this dataset's approval for group ML_1371_12321_2

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Dataset Description

This dataset has customers records from a telecommunications company. Each row is a customer of the company, and the rows describe different things about the customers like the customer's account or the amount of service usage. We would like to focus on the churn column, which states whether the customer left the company recently or not.

Proposal

We want to use this data set to build a machine learning model that can predict if the customer will churn or not. We want to be able to predict this as losing customers in the telecom providers lose a lot of money and they lose a long term customer. If you were able to flag this a month before it happened, you may give the customer extra benefits if they sign a contract with you, keeping the customer for a longer period of time. This would let the company understand their weak points, as they can see any customer about to leave and ask about the problems they have, and the company can take action to fix them.

Over the course, our group will work together to 1. Clean and prepare the data set to be ready for training. 2. Perform EDA to understand our data and make correct predictions on who will leave based on other columns. 3. Encode categorical variables to make everything number based inside the dataset, so turning partner from yes or no to 1 or 0. Scale numerical features to all be on the same scale and get rid of class imbalances using `class_weight='balanced'`. 4. We then want to train and compare different models like logistic regression or the decision tree to see the best one. 5. Finally put our models to the test and test their accuracy and recall.